CHE 321 – Chemical Reaction Engineering
Fall 2019

Homework Assignment – 3 (40 points)

Due: 12 PM, October 21st 2019

Problem-1: Derivation of Rate Expression from the Reaction Mechanism (20 points)

In class lecture, we have learned how to derive a rate expression using **pseudo-steady state approximation**. According to this approximation, the rate of change of concentrations of the intermediate species are zero i.e., the concentrations of the intermediate species do not change significantly with time as the reaction progresses.

A) Nitrogen Pentoxide Decomposition

Consider the decomposition of nitrogen pentoxide into nitrogen dioxide and oxygen,

$$\mathbf{N_2O_5 \rightarrow 2\,NO_2 + \tfrac{1}{2}\,O_2} \qquad \textbf{R1}$$

This is not an elementary reaction and its mechanism consists of the following set of elementary reactions.

$$\mathbf{N_2O_5 \rightarrow NO_3 + NO_2} \qquad \textbf{R1a}$$

$$\mathbf{NO_2 + NO_3 \rightarrow N_2O_5} \qquad \textbf{R1b}$$

$$\mathbf{NO_2 + NO_3 \rightarrow NO_2 + O_2 + NO} \qquad \textbf{R1c}$$

$$\mathbf{NO + NO_3 \rightarrow 2NO_2} \qquad \textbf{R1d}$$

The rate constants for the reactions R1a, R1b, R1c and R1d are denoted as k_1, k_2, k_3 and k_4, respectively. Here, NO and NO_3 are regarded as the intermediate species.

(a) Derive the rate expression for the overall reaction R1 from its mechanism (R1a-d) using the pseudo- steady state approximation. (**8 points)**

(b) What is the overall order of the reaction? (**1 point**)

(c) Based on the order of the reaction R1 what do you infer? (**1 point**) *(Hint: Look at the reaction stoichiometry of R1)*

B) Hydrogen Bromide Formation

Consider the formation of hydrogen bromide given by,

$$\mathbf{\tfrac{1}{2}\,H_2 + \tfrac{1}{2}\,Br_2 \rightarrow HBr} \qquad \textbf{R2}$$

It is not an elementary reaction and it's mechanism can be written as

$$Br_2 \rightarrow 2Br^* \qquad \textbf{R2a}$$

$$Br^* + H_2 \rightarrow H^* + HBr \qquad \textbf{R2b}$$

$$H^* + Br_2 \rightarrow Br^* + HBr \qquad \textbf{R2c}$$

$$H^* + HBr \rightarrow H_2 + Br^* \qquad \textbf{R2d}$$

$$2Br^* \rightarrow Br_2 \qquad \textbf{R2e}$$

The rate constants for the reactions R2a, R2b, R2c, R2d and R2e are denoted as k_1, k_2, k_3, k_4 and k_5, respectively. Derive the rate expression for R2 from its mechanism using the pseudo-steady state approximation. (**10 points**)

Problem-2: Biochemical Reaction with Lag Phase

A lag phase in the growth curve is typically observed when fresh biomass is mixed with the nutrients (see Fig 1). The biomass population during this lag phase stays the same. This lag phase could be because of the absence of a specific enzyme that is required so that the microorganisms can utilize a specific substrate. The Monod based growth rate is incapable of predicting this lag phase.

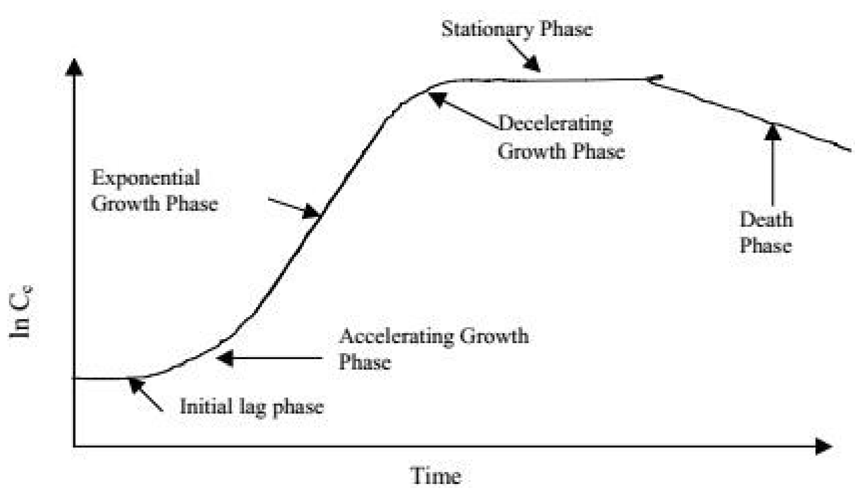

Fig 1. Typical growth phases in a batch culture

A) Consider the following reaction:

$$\mathbf{B + 2S_1 \rightarrow 2B}$$

Plot ln C_B vs. time for upto 8 hours when the initial concentrations for S_1 and the biomass B are 4 and 0.13 g/l, respectively. Take μ_1 = 0.9 hr^{-1} and Ks_1 = 0.1 g/l .

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[Hint: you should solve the differential equation of the substrate and biomass simulatanously.] ***(10 points)***

To correct the Monod growth rate expresson for lag phase, we introduce the rate-limiting enzyme concentration to the equation as follows.

$$r = \frac{\mu_{max} E_R C_S}{K_S + C_S} C_B$$

where E_R is the relative Enzyme ratio given as

$$E_R = \frac{e}{e_{max}}$$

and defined as the current enzyme content to its maximum value inside the cell. The unit of *e* is gm of enzyme/gm of cell. e_{max} is defined as

$$e_{max} = \frac{\alpha}{\mu_{max} + \beta}$$

where α and β are enzyme synthesis and degredation rate constants. Then the balance equation

for the system can be written as

$$\frac{d(eC_B)}{dt} = \frac{\alpha C_S C_B}{K_S + C_S} - \beta(eC_B)$$

$$\frac{d(e)}{dt} = \frac{\alpha C_S}{K_S + C_S} - \beta(e) - \frac{1}{C_B}\frac{dC_B}{dt} e$$

This equation along with substrate and biomass differential equations can be simulated to produce the necessary curve.

B) Plot C_B vs. time for the question A when the rate limiting enzyme concentration is 8.3 x10^{-5} g/l initially. Consider μ_1 =0.9 hr^{-1}, K_{s1}= 0.1 g/l, $\alpha_{1,2}$=0.0001 hr^{-1} and $\beta_{1,2}$= 0.05 hr^{-1} (10 points)

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Now we want to try the same thing when we have two substrates in the system.

Consider the cell growth in a batch reactor with two different substrates S_1 and S_2 . we will have:

$$B + \alpha S_i \rightarrow 2B \quad i = 1.2$$

With the reaction rates:

$$r = \frac{\mu_i C_{Si} E_i C_B}{K_{Si} + C_{Si}} \quad i = 1.2$$

In the case of multiple substrates, usually one substrate utilization happens at the time. This means that the microorganisms utilize S_1 and only after the complete utilization of S_1, S_2 is being consumed. So, in other words, the r is going to be either r_1 or r_2 at a specific time.

The sequential consumption results in a pattern in which the cells consume the preferred substrate and then after an intermediate lag phase, they utilize the second substrate. This is called diauxic growth. You can see the growth curve below.

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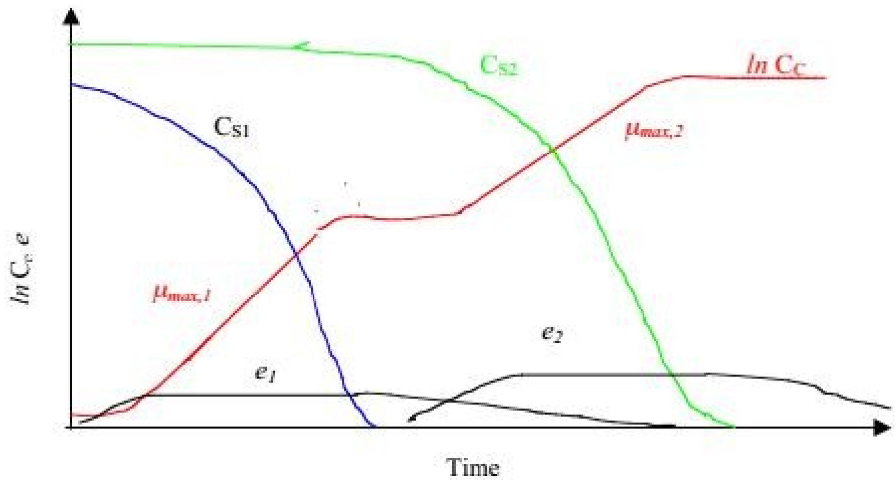

Figure 3. Diauxic growth of bacteria on two substrates, showing also the intracellular content of the two hypothetical key enzymes for consuming each substrate.

To model the diauxic growth with modified Monod equatios discussed above, the concept of cybernetic variables is introduced. These variables u_i and v_i represent the outcome of microbial processes controlling enzyme utilization and substrate consumption. The differential equation for this system can be written as follows:

$$\frac{dC_B}{dt} = (\mu_1 v_1 + \mu_2 v_2)C_B$$

$$\mu_i = \frac{\mu_{maxi} C_{Si}}{K_{Si} + C_{Si}} \quad i = 1.2$$

$$\frac{dC_{S1}}{dt} = \frac{-1}{\alpha_{S1}} \mu_1 v_1 C_C$$

$$\frac{dC_{S2}}{dt} = \frac{-1}{\alpha_{S2}} \mu_2 v_2 C_C$$

$$\frac{d(e_i)}{dt} = \frac{\acute{\alpha}_i C_{Si}}{K_{Si} + C_{Si}} u_i - \beta_i e_i - \frac{1}{C_C}\frac{dC_C}{dt} e_i \quad i = 1.2$$

where the cybernetic variables are

$$\mu_i = \frac{\mu_i}{\mu_1 + \mu_2} \quad and\ v_i = \frac{\mu_i}{max\mu_j} \qquad i\&j = 1.2$$

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C) (Bonus question) Assume we have the following reactions:

$$\mathbf{B + 2S_1 \rightarrow 2B}$$

$$\mathbf{B + 3S_2 \rightarrow 2B}$$

the initial concentrations for S_1, S_2 the biomass and the enzymes are 4, 20 and 0.13, $e_{1,10}$=8.3x10^{-5} and $e_{2,0}$=10^{-6} g/l ,respectively. μ_1 = 0.9 hr^{-1} , μ_2= 0.6 hr^{-1} , Ks_1= 0.1 g/l , Ks_2= 0.5 g/l ,$\alpha_{1,2}$=0.0001 hr^{-1} and $\beta_{1,2}$= 0.05 hr^{-1}

plot $\ln C_B$ vs time in the matlab for time span of 0 to 3 hours. (10 points)